Generative AI and LLM

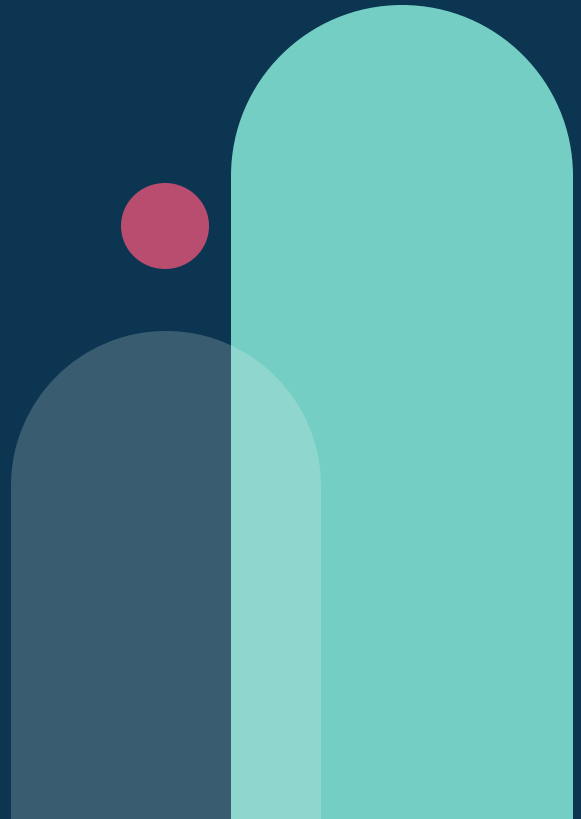
Generative AI

Basics of LLMs

Popular LLMs

01

Generative AI



Introduction to Generative AI

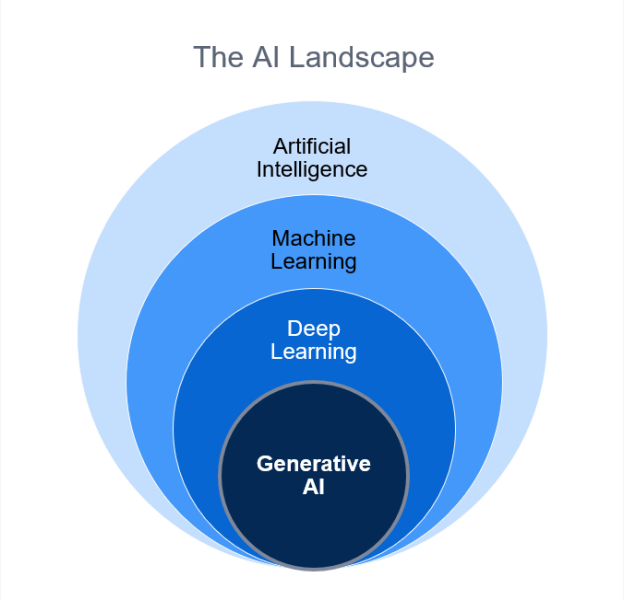
- **Definition**: Generative AI refers to a type of artificial intelligence that can create new, unique content such as artwork, music, text, and even realistic images.
- **Comparison with Traditional AI**: While traditional AI focuses on specific tasks and problem-solving, generative AI is characterized by its ability to exhibit creativity similar to that of humans.
- **Human-like Creativity**: Generative AI is capable of producing new ideas, content, or solutions, drawing on its learned knowledge to create something entirely novel.

Generative AI in the AI Hierarchy

AI Hierarchy Overview:

- **Artificial Intelligence (AI)**: Broad field focused on creating systems that can perform tasks that typically require human intelligence.
- **Machine Learning (ML)**: Subset of AI that uses data and algorithms to enable machines to improve their performance over time.
- **Deep Learning**: A subset of ML that uses neural networks with multiple layers to learn from vast amounts of data.
- **Generative AI**: Positioned as a subset of deep learning, distinguished by its focus on creating rather than just predicting or classifying.

Key Differentiator: Unlike predictive models that classify or identify patterns, generative AI can generate entirely new content, showcasing its creative capabilities.



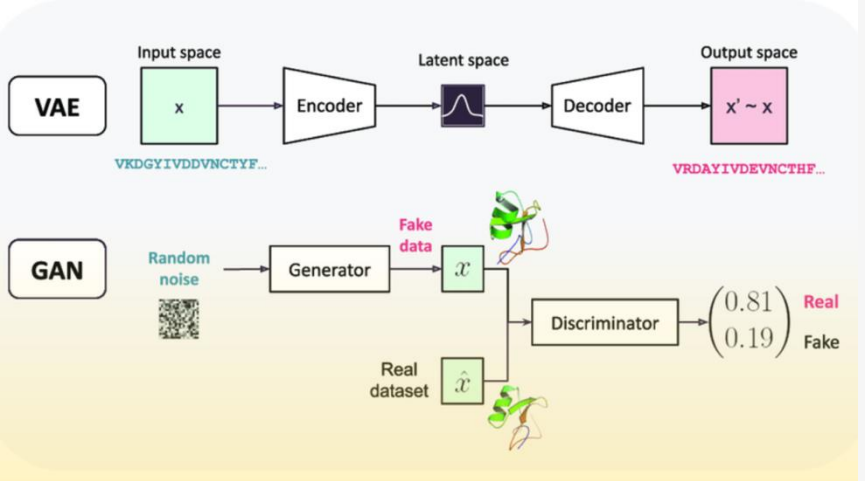
Generative Models

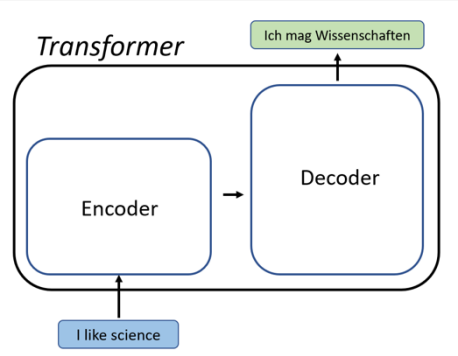
Definition: Generative models are a class of machine learning models that learn the underlying distribution of data and can generate new data that resembles the original data.

Types of Generative Models:

- **Variational Autoencoders (VAEs)**: Generate new data by learning the latent space of the input data.
- **Generative Adversarial Networks (GANs)**: Consist of two neural networks (a generator and a discriminator) that work together to produce realistic data.
- **Transformer-based Models**: Utilized in natural language processing for generating coherent text or other sequential data.

Applications: These models are widely used for creating images, text, music, and other forms of media that closely mimic human creativity.





Text Generation

Overview: Text generation is one of the most common applications of generative AI, enabling the creation of new textual content.

Tools: ChatGPT and similar models are prominent examples of generative AI tools used for text generation.

Applications:

- **Content Creation**: Articles, reports, poetry, stories, and other forms of written content.
- **Virtual Assistants and Chatbots**: These tools generate contextually relevant and coherent responses in natural language, mimicking human conversation.

Significance: The ability to generate human-like text has revolutionized content creation and automated customer interactions.



Image Generation

Overview: Generative AI models can generate new, creative images by learning from large datasets of existing images.

Prominent Tools:

- **DALL-E 3**
- **Flux**
- **Midjourney**
- **Adobe Firefly**
- **Stable Diffusion**

Applications:

- **Image-to-Image Translation**: Transforming an image from one style to another.
- **Text-to-Image Translation**: Creating images based on textual descriptions.
- **Creative Art**: Generating entirely new artistic expressions based on learned patterns.

Impact: These tools are increasingly used in art, advertising, and media production, offering new possibilities for creative expression.



Video and Audio Generation

Video Generation:

Definition: Generative models can create entire videos from scratch by stitching together scenes, characters, and actions.

Applications: Used in entertainment, advertisements, and training simulations. Video game development is a significant area of application.

Advanced Capabilities: Some models can predict and generate future video frames, useful for scenarios like repairing damaged video footage.

Video Generation Tools:

- **Runway**
- **HeyGen**
- **Pika Labs**
- **Sora**

Audio Generation

Definition: Generative AI can analyze audio data to mimic or generate entirely new voices.

Applications: Virtual assistants, audiobooks, and voiceover work, making interactions sound more natural and personalized.

Audio Generation Tools:

- **StableAudio**
- **elevenlabs**
- **AudioCraft**
- **Speechify**

Healthcare Applications

Synthetic Data Generation:

- **Definition**: Generative AI can create synthetic data that closely resembles real-world data, useful in fields like healthcare where data availability may be limited.
- **Applications**:
  - **Medical Research**: Generating synthetic patient data for research without compromising patient privacy.
  - **Drug Discovery**: Accelerating drug development by generating new molecular structures with desired properties.

Significance: These capabilities are transforming healthcare by providing new tools for research, reducing costs, and accelerating innovation.

Gaming and Art Generation

Gaming:

- **Impact**: Generative AI is revolutionizing the gaming industry by automating the creation of complex, immersive worlds, characters, and storylines.
- **Applications**:
  - **World-Building**: Creating vast, detailed virtual environments.
  - **Character Creation**: Designing unique characters with distinct traits and personalities.

Art Generation:

- **Creativity**: Generative AI can produce original artwork, including paintings, poetry, and multimedia art.

Significance: This extends the boundaries of human creativity, offering new forms of artistic expression.

Software Development and Finance

Software Development:

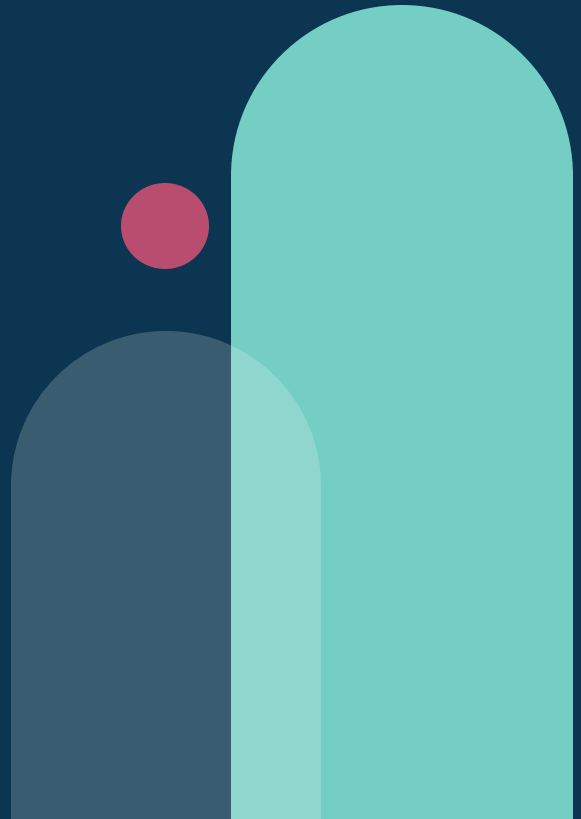
- **Tools**:
  - **GitHub Copilot**
  - **AlphaCode**
  - **Devin**
- **Applications**: Accelerating development cycles, enhancing software testing, and improving code quality.

Finance:

- **Impact**: Financial institutions use generative AI for market analysis, risk assessment, and trading strategy optimization.
- **Applications**: Fraud detection, portfolio optimization, and improving investment decisions through predictive insights.

02

LLMs



Introduction to LLMs

Definition:

LLMs are a type of AI model designed to understand and generate human-like text by processing large datasets.

Core Functionality:

These models can generate coherent text, translate languages, summarize information, and answer questions.

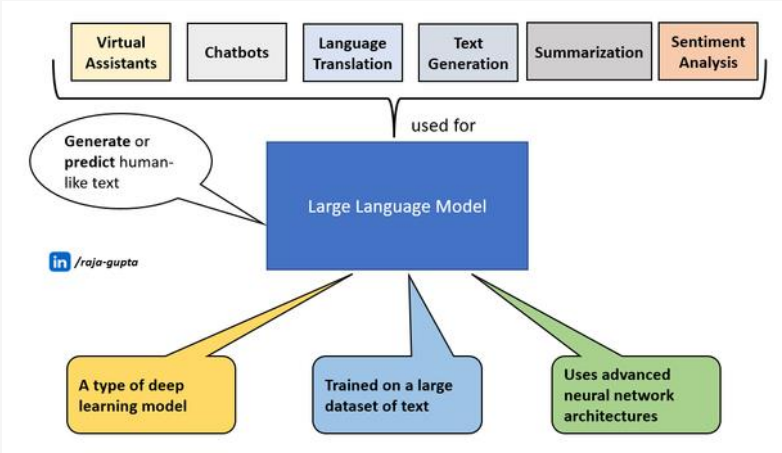
Capabilities of LLMs

Text Generation: Producing human-like text based on prompts.

Language Translation: Translating text between languages while preserving meaning.

Summarization: Condensing long texts into concise summaries.

Question Answering: Responding to queries with relevant and coherent answers.

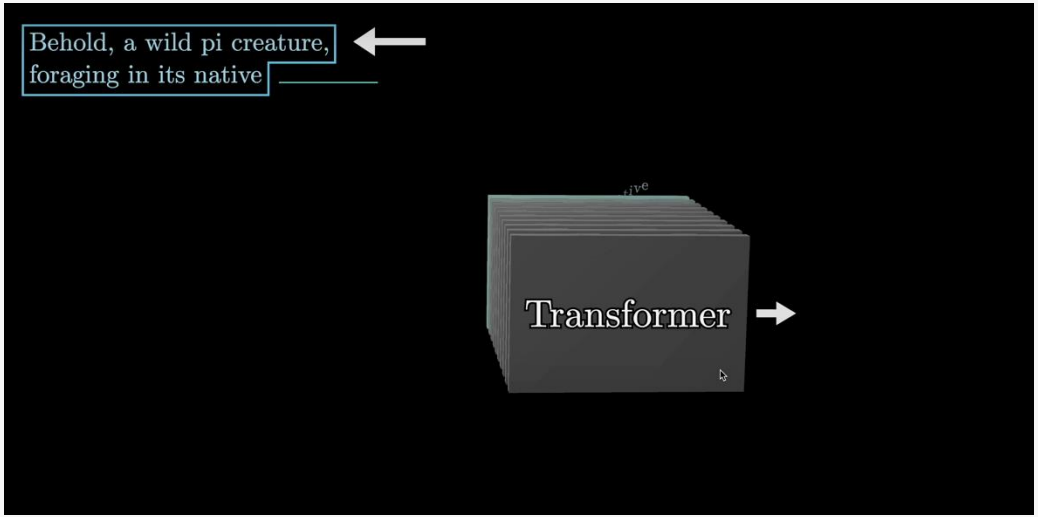


How LLMs Work

Architecture: Typically based on transformer models, which excel at processing sequential data.

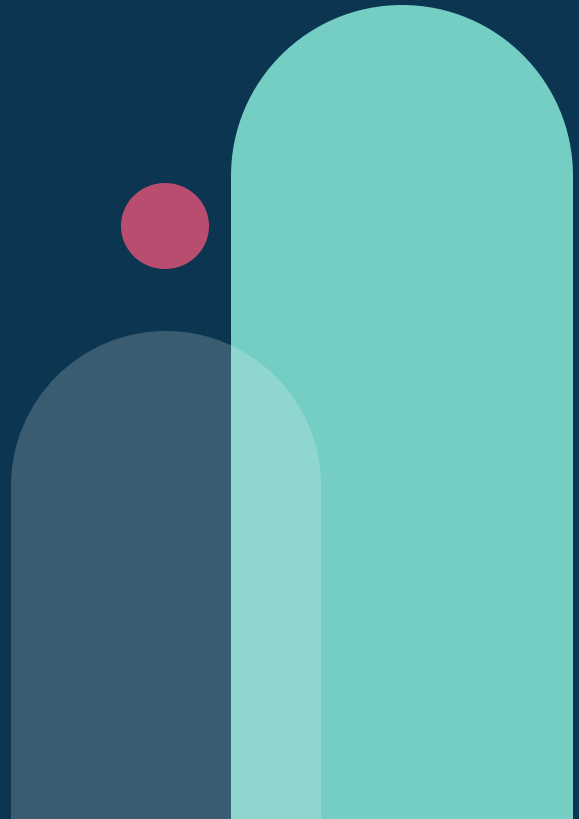
Training: LLMs are trained on vast amounts of text data, learning patterns, structures, and meanings from diverse sources.

Output: Generates text by predicting the next word in a sequence, informed by its training data.



03

Popular LLM Architectures



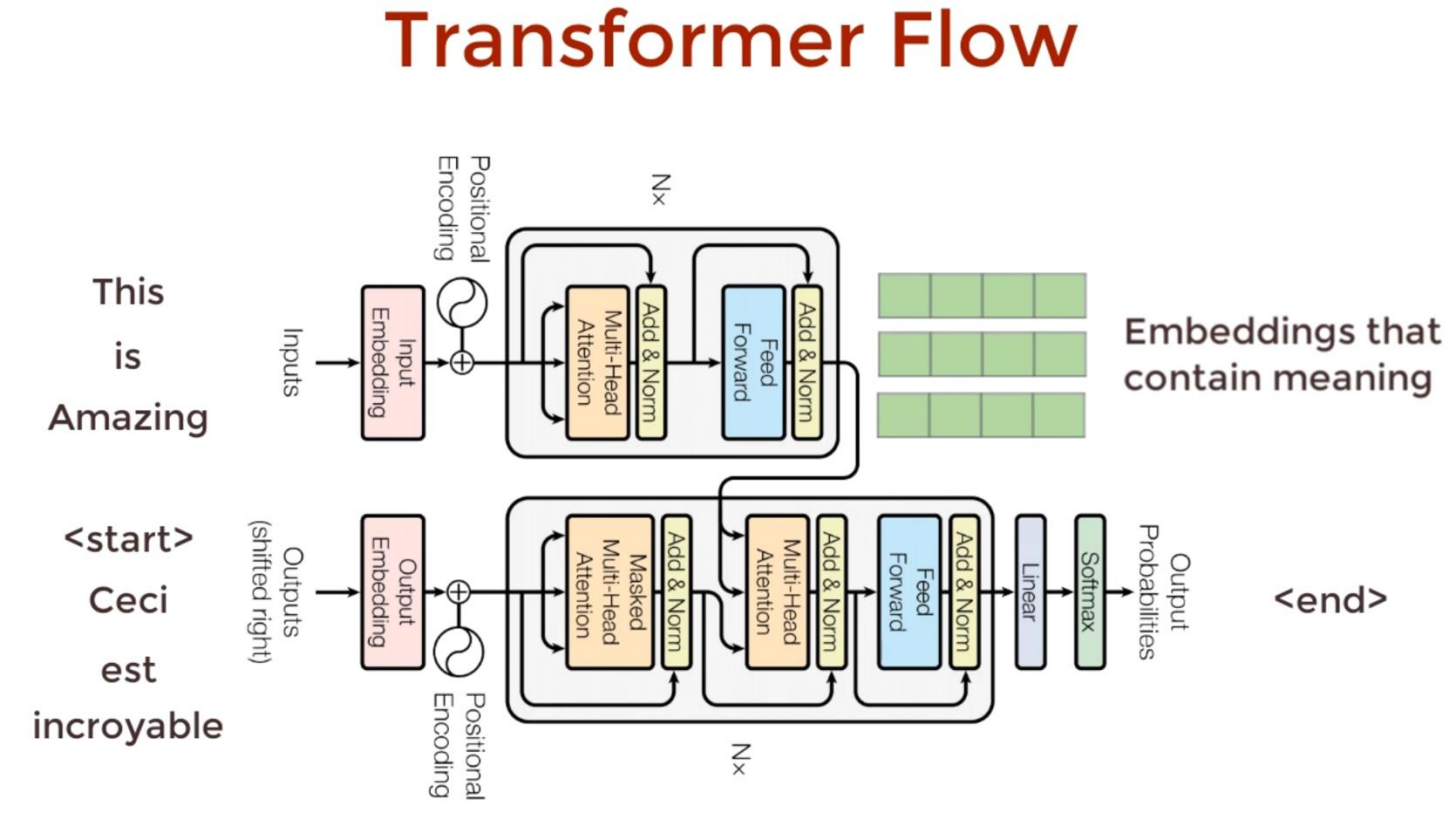
Evolution of LLMs

Historical Context: The progression from simple statistical models to complex neural networks and transformers.

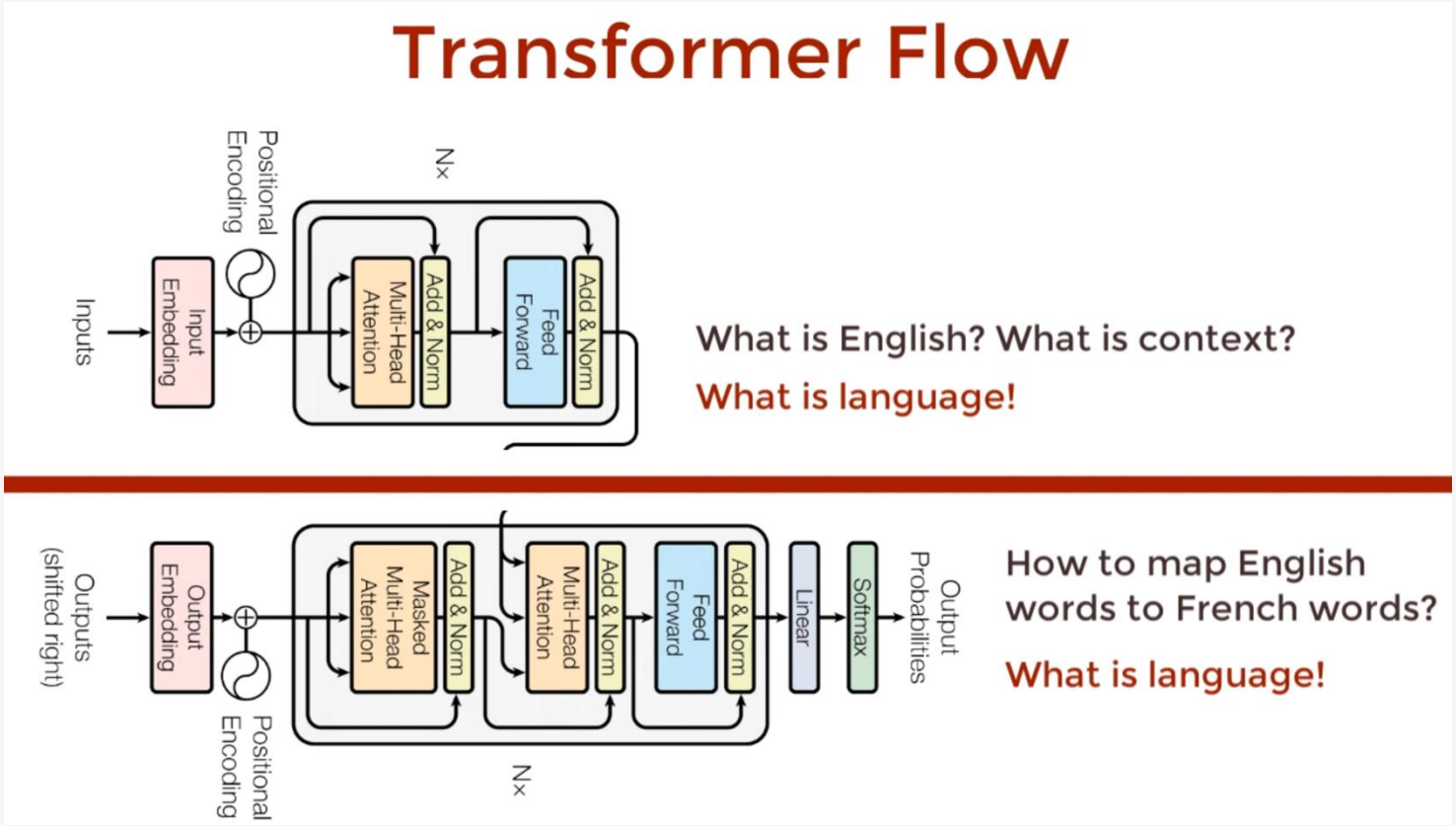
Notable Models:

- **GPT (Generative Pre-trained Transformer)**: One of the earliest and most influential LLMs.
- **BERT (Bidirectional Encoder Representations from Transformers)**: Known for understanding context by looking at words from both directions.

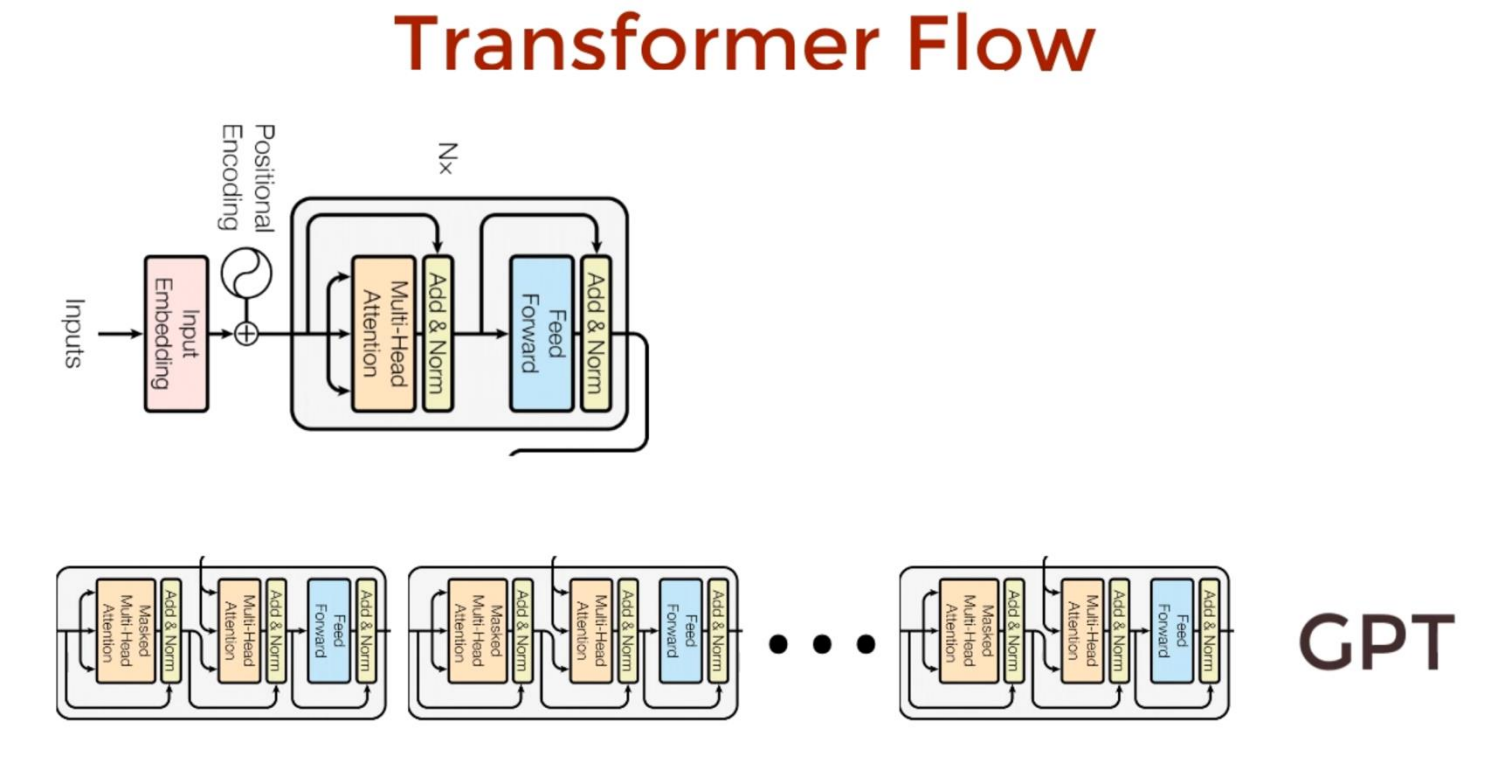
Evolution of LLMs



Evolution of LLMs



Evolution of LLMs



GPT-2

Generative

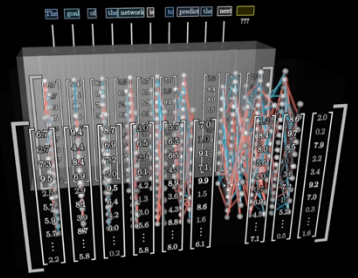
computer science is to practice coding regularly. Start by working on simple exercises and gradually move on to more complex projects. 4. Participate in coding challenges and competitions: Coding challenges and competitions provide a great opportunity to put your skills to the test and learn from

Pre-trained

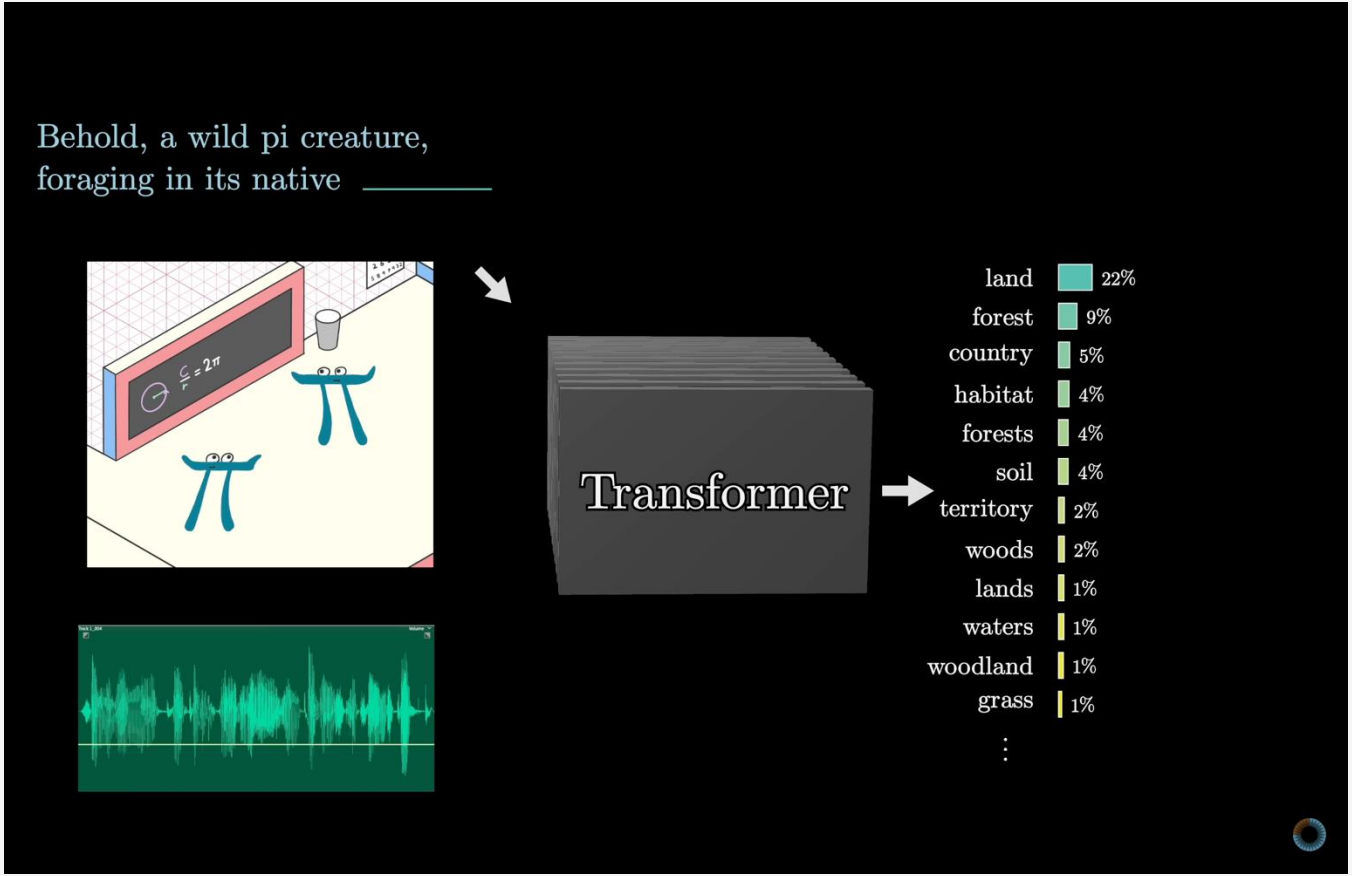
the gentle Mr. Lorry. “I am not crying,” said Miss Pross; “_you_ are.” “I, my Pross?” (By this time, Mr. Lorry dared to be pleasant with her, on occasion.) “You were, just now; I saw you do it, and I don’t wonder at it. Such a present of plate as



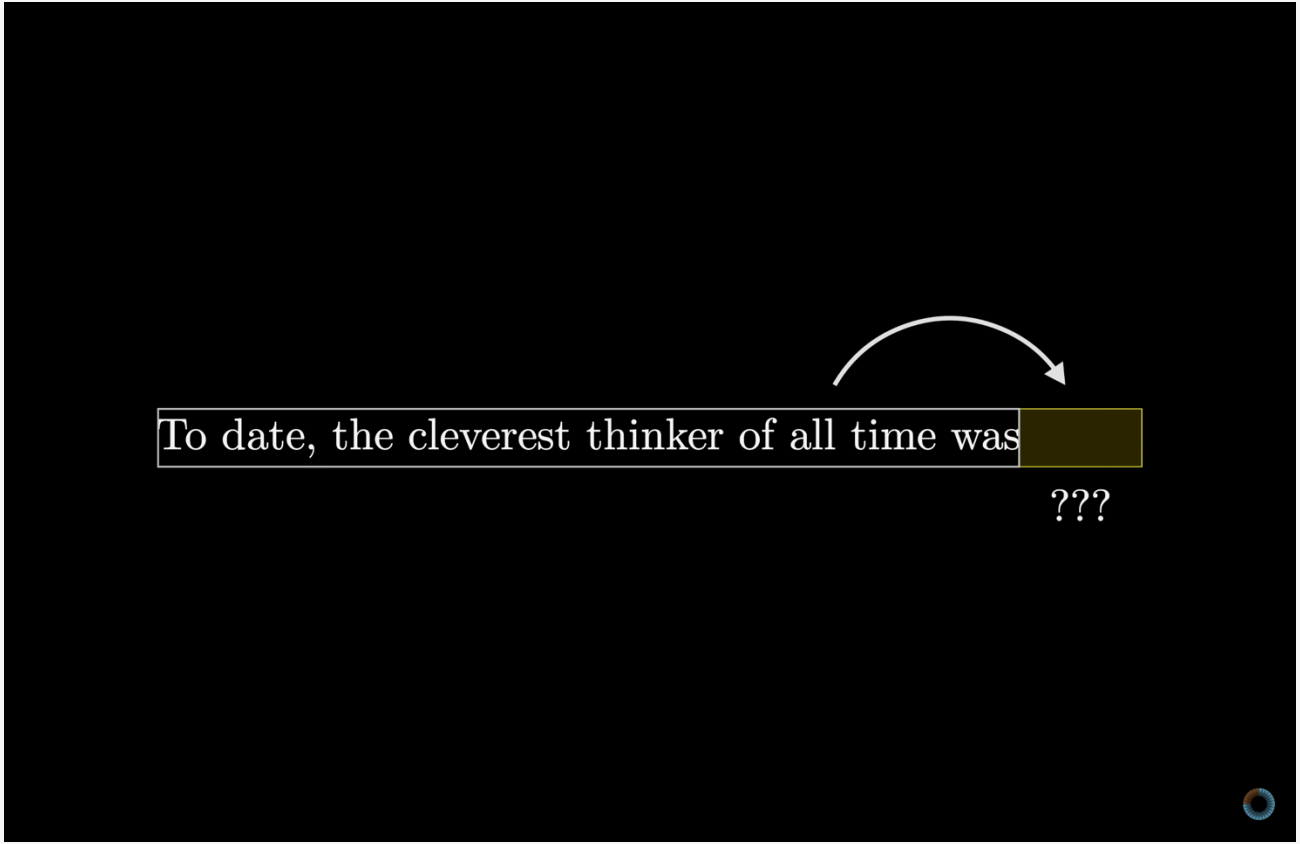
Transformer



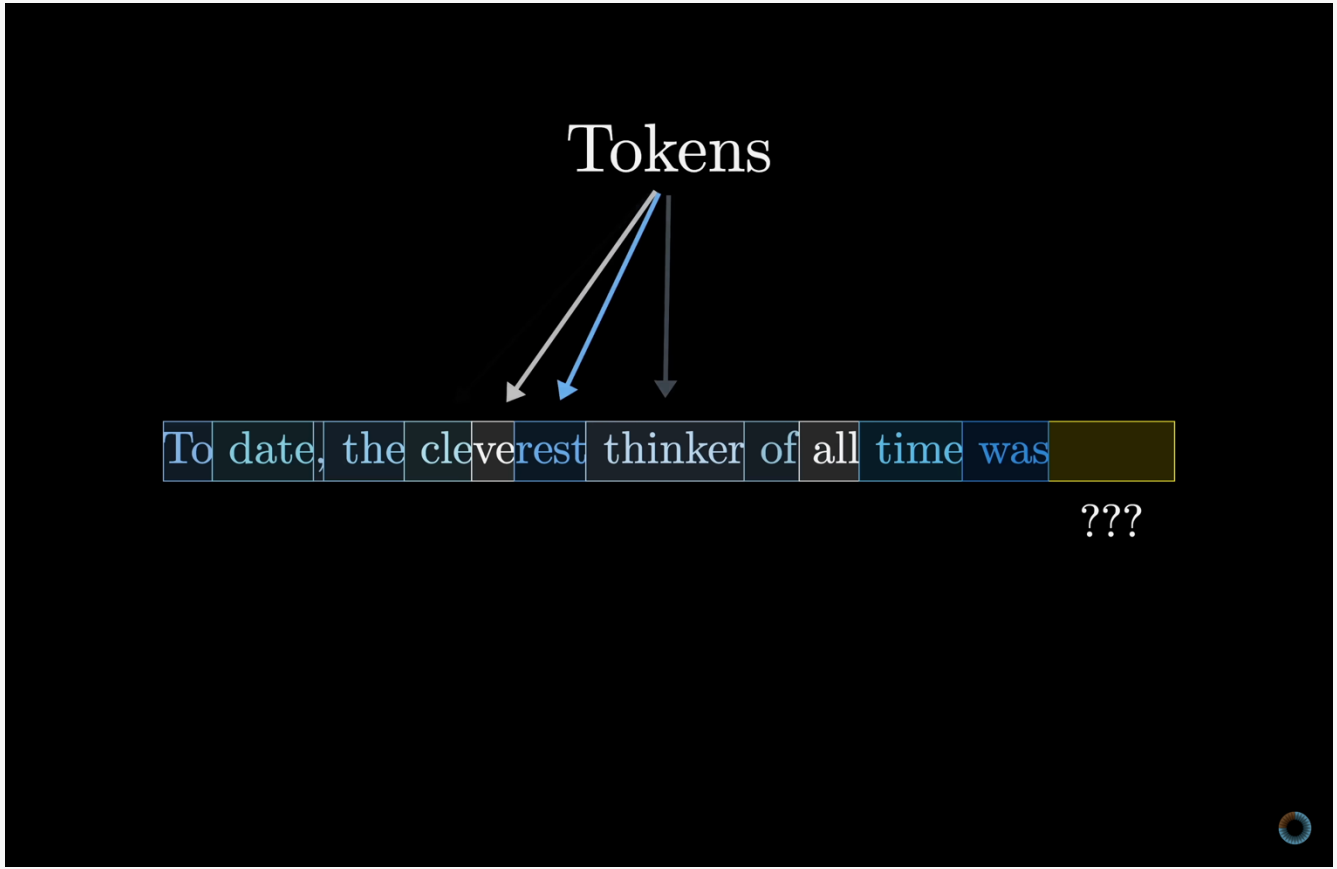
GPT-2


Behold, a wild pi creature,
foraging in its native _____
Transformer
land 22%
forest 9%
country 5%
habitat 4%
forests 4%
soil 4%
territory 2%
woods 2%
lands 1%
waters 1%
woodland 1%
grass 1%

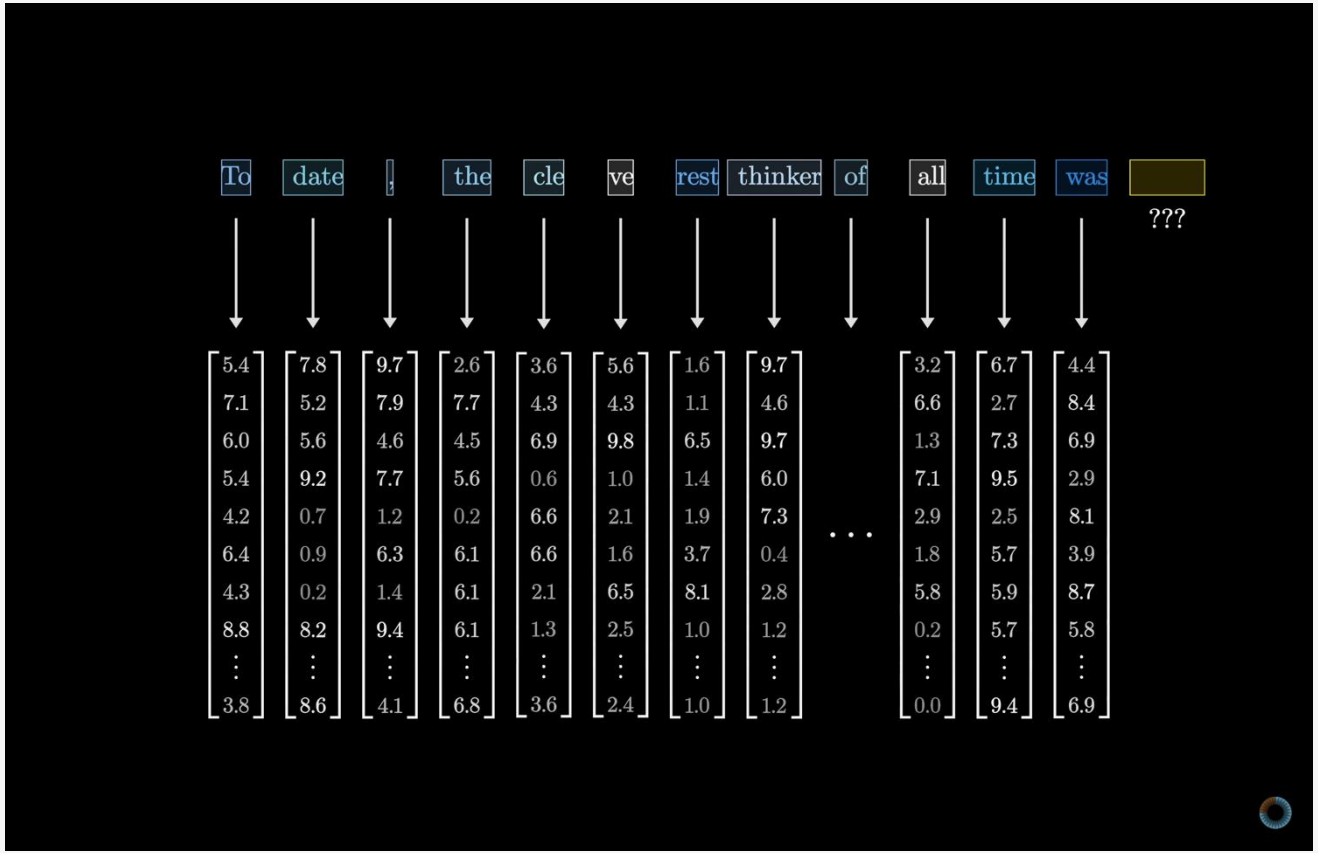
Next Word Prediction



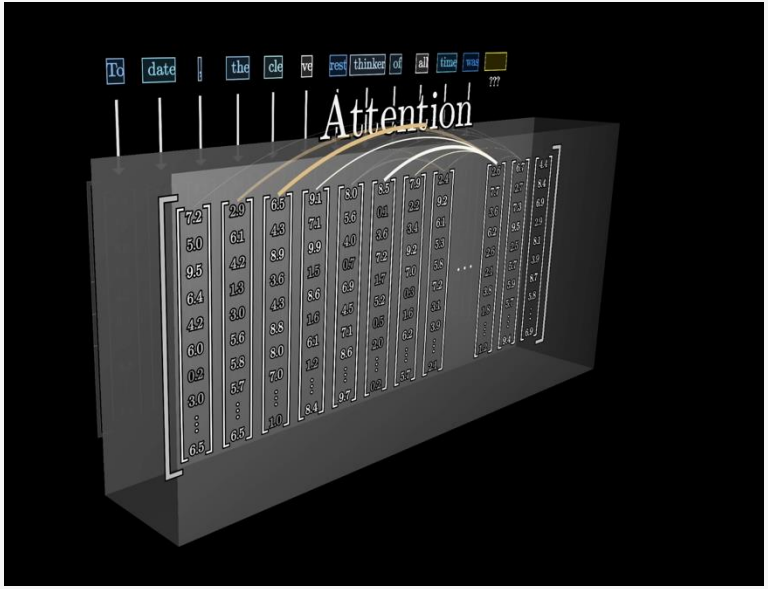
Tokenization

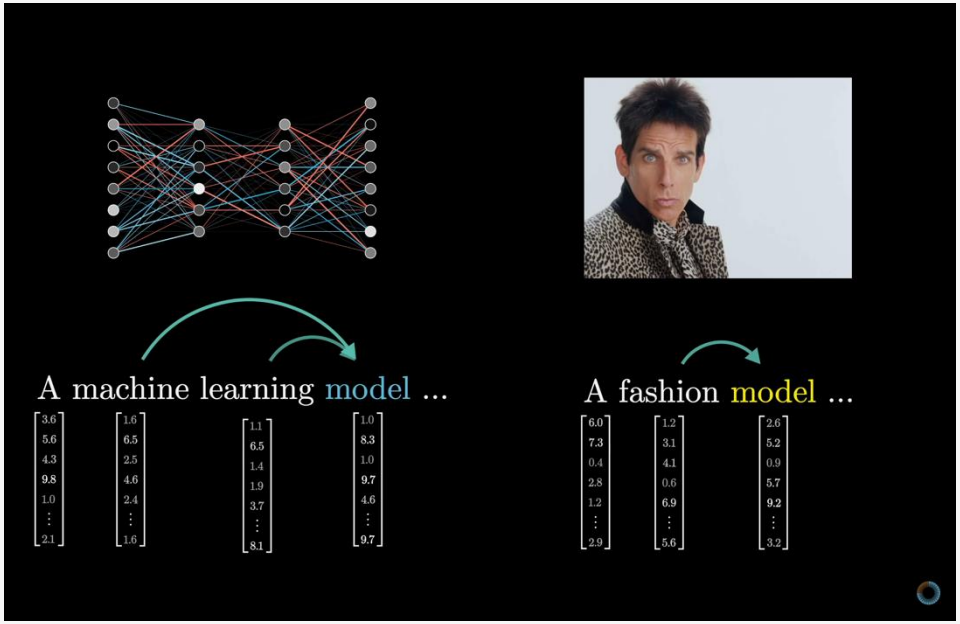


Embeddings

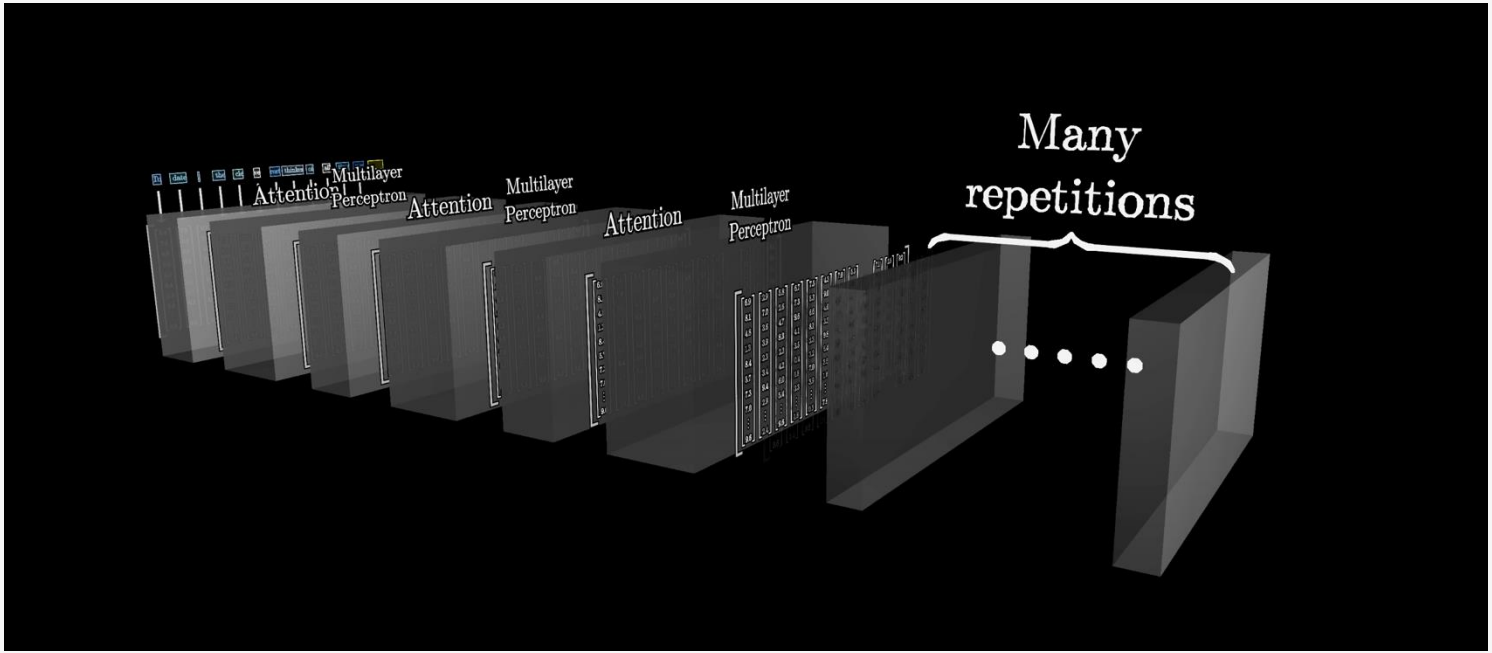

To
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thinker
of
all
time
was
???
5.4 7.1 6.0 5.4 4.2 6.4 4.3 8.8 3.8
7.8 5.2 5.6 9.2 0.7 0.9 0.2 8.2 8.6
9.7 7.9 4.6 7.7 1.2 6.3 1.4 9.4 4.1
2.6 7.7 4.5 5.6 0.2 6.1 6.1 6.1 6.8
3.6 4.3 6.9 0.6 6.6 6.6 2.1 1.3 3.6
5.6 4.3 9.8 1.0 2.1 1.6 6.5 2.5 2.4
1.6 1.1 6.5 1.4 1.9 3.7 8.1 1.0 1.0
9.7 4.6 9.7 6.0 7.3 0.4 2.8 1.2 1.2
3.2 6.6 1.3 7.1 2.9 1.8 5.8 0.2 0.0
6.7 2.7 7.3 9.5 2.5 5.7 5.9 5.7 9.4
4.4 8.4 6.9 2.9 8.1 3.9 8.7 5.8 6.9

Attention

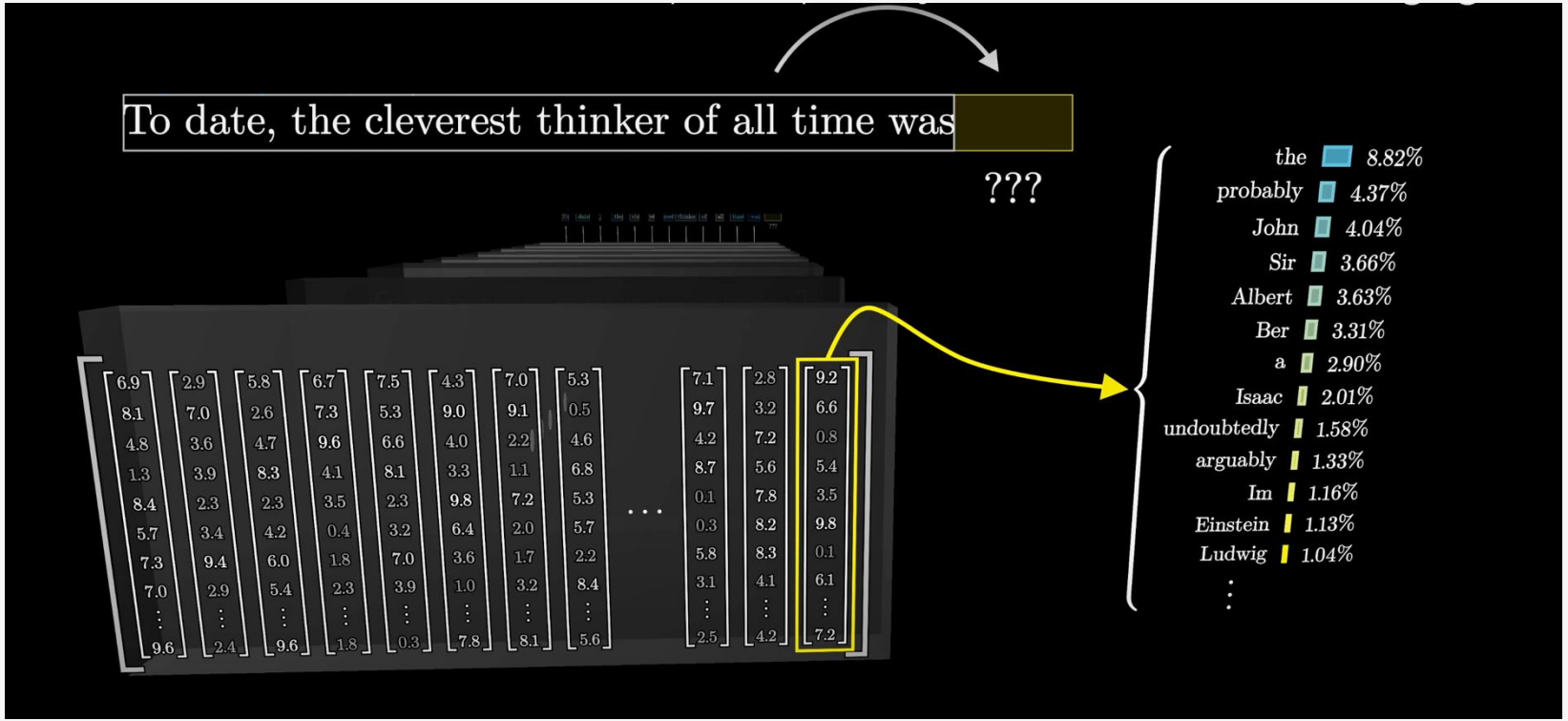

To
date
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thinker
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???
Attention


A machine learning model ...
A fashion model ...

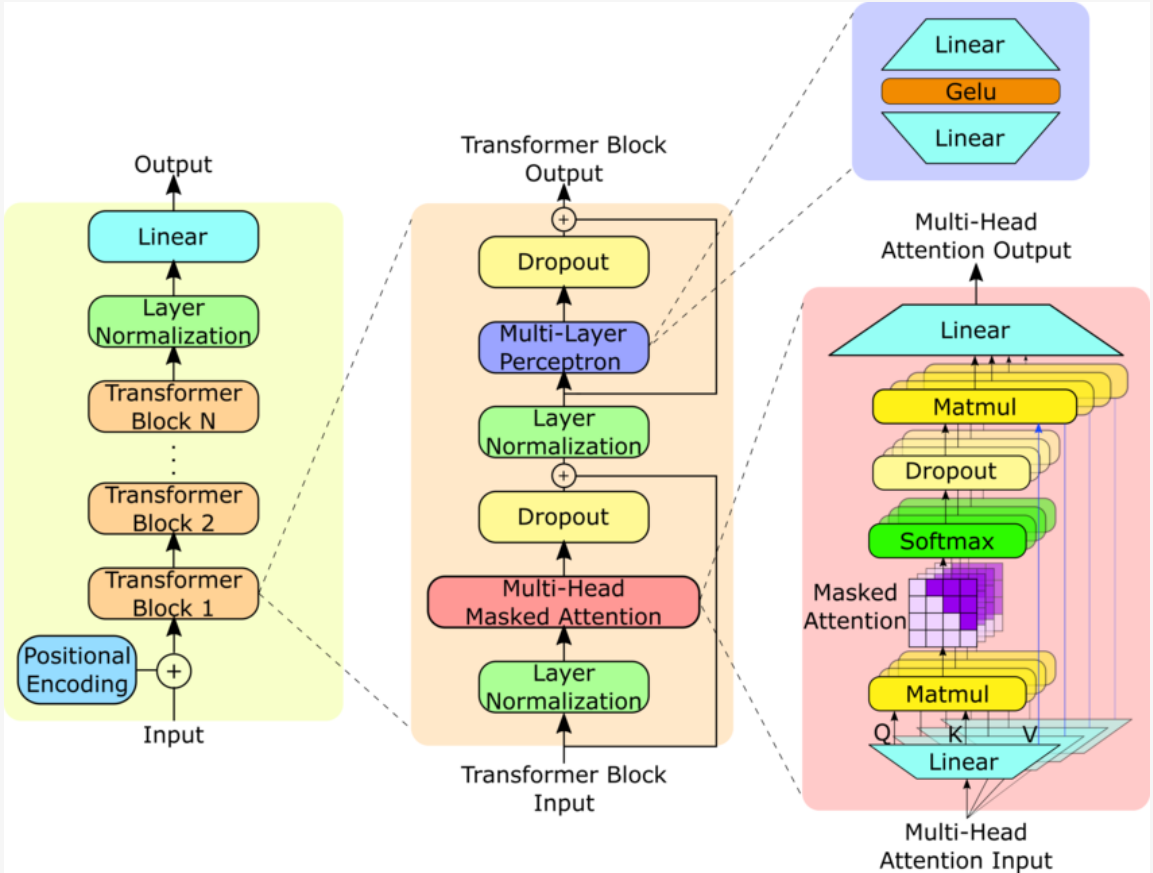
Architecture



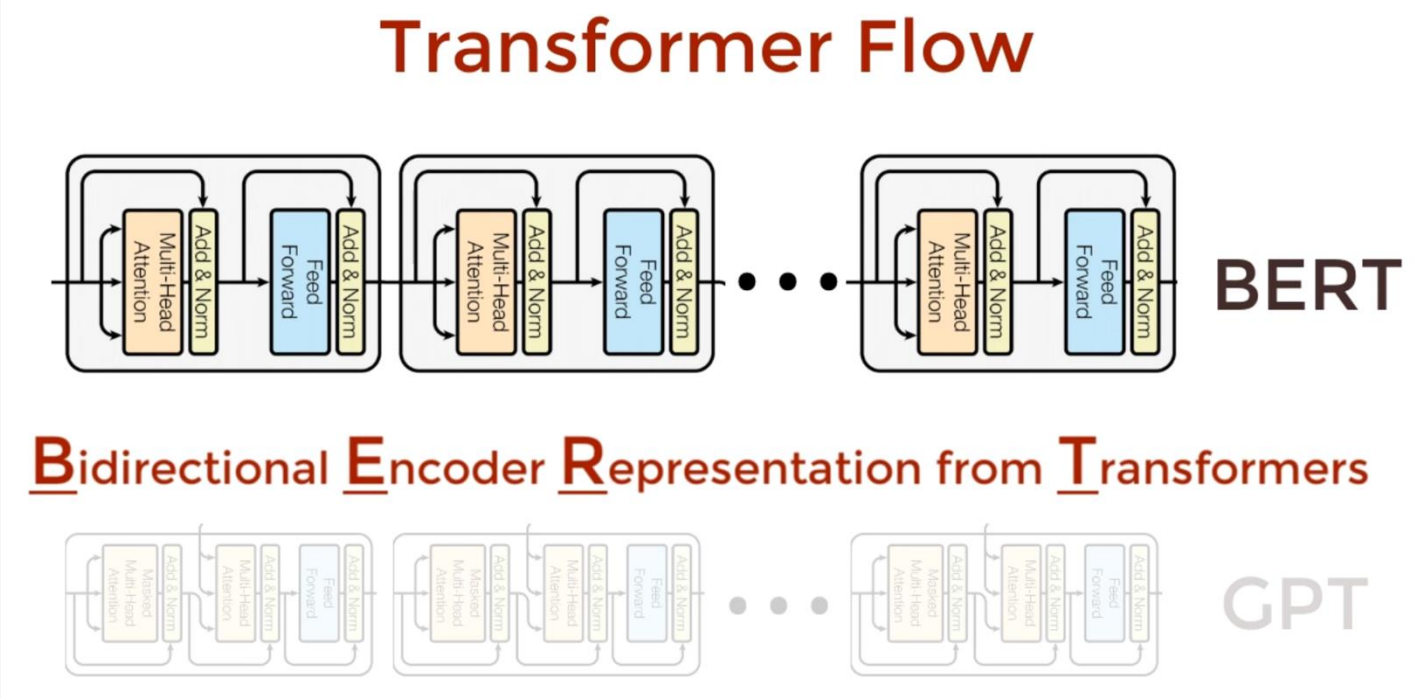
Outputs


To date, the cleverest thinker of all time was
???
the 8.82%
probably 4.37%
John 4.04%
Sir 3.66%
Albert 3.63%
Ber 3.31%
a 2.90%
Isaac 2.01%
undoubtedly 1.58%
arguably 1.33%
Im 1.16%
Einstein 1.13%
Ludwig 1.04%

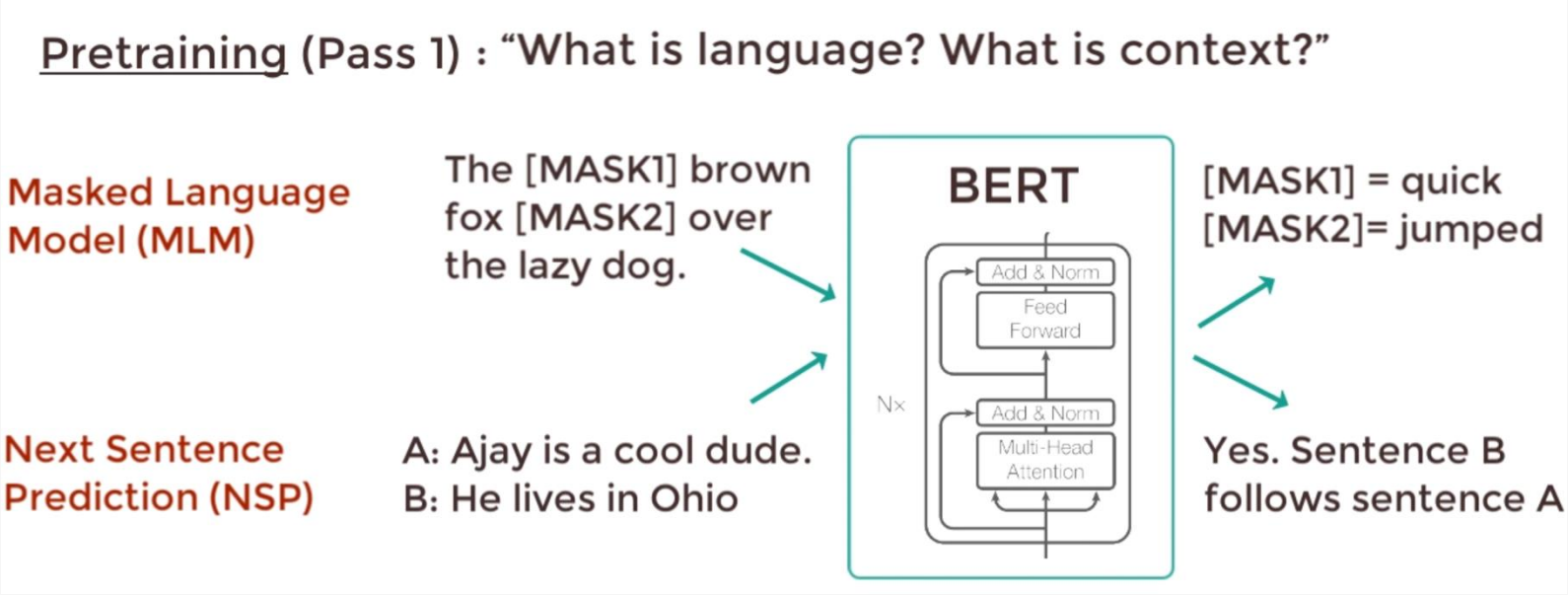
GPT-2


Output
Linear
Layer Normalization
Transformer Block N
Transformer Block 2
Transformer Block 1
Positional Encoding
Input
Transformer Block Output
Dropout
Multi-Layer Perceptron
Layer Normalization
Dropout
Multi-Head Masked Attention
Layer Normalization
Transformer Block Input
Linear
Gelu
Linear
Multi-Head Attention Output
Linear
Matmul
Dropout
Softmax
Masked Attention
Matmul
Q
K
V
Linear
Multi-Head Attention Input

BERT

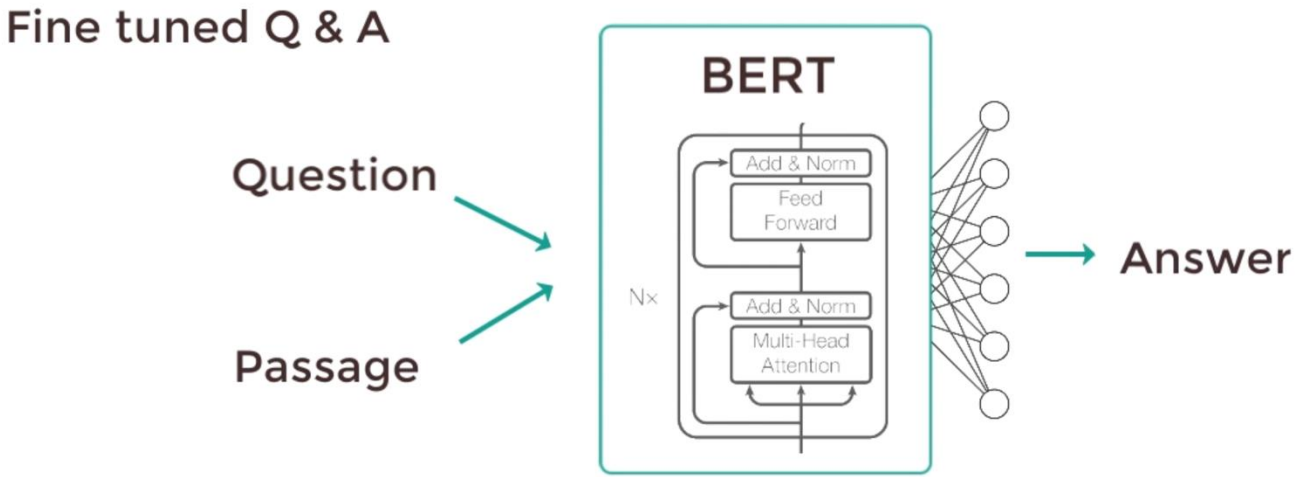

Transformer Flow
Multi-Head Attention
Add & Norm
Feed Forward
Add & Norm
BERT
Bidirectional Encoder Representation from Transformers
GPT

Pre-Training BERT



Fine-Tuning BERT

Fine Tuning (Pass 1): “How to use language for specific task?”



Applications of LLMs

Content Creation: Generating articles, reports, and creative writing.

Virtual Assistants: Powering chatbots and AI-driven customer support.

Code Generation: Assisting in programming by generating code snippets and entire functions.

Education: Providing personalized tutoring and language learning support.

Challenges and Limitations

Bias: LLMs can reflect and amplify biases present in the training data.

Computation: Training and running LLMs require significant computational resources.

Accuracy: While powerful, LLMs can sometimes generate incorrect or nonsensical information.

Ethical Considerations and Future

Ethical Considerations:

- **Misinformation**: Potential for LLMs to generate false or misleading content.
- **Privacy**: Concerns over the data used to train these models and the potential for privacy breaches.
- **Responsibility**: Importance of using LLMs responsibly to avoid harmful applications.

Future of LLMs:

- **Continued Improvement**: Ongoing research to make LLMs more efficient, accurate, and ethical.
- **Integration**: Greater integration into everyday applications across industries.
- **Democratization**: Efforts to make LLM technology more accessible to developers and organizations.

Summary and Impact

Summary: LLMs represent a significant advancement in AI, enabling machines to understand and generate human-like text.

Impact: Transforming industries by automating tasks, enhancing communication, and driving innovation in AI applications.

Thanks!

Do you have any questions?

